WALMART CUSTOMER RETENTION ANALYTICS

POWER BI PROJECT REPORT

Submitted By: **Athul Santo**

Course: **Data Science**

Submission Date: **14-01-2004**

Table of Contents

Introduction ---------- 3

Dataset description ---------- 4

Data cleaning & modeling ---------- 5

DAX measures used ---------- 6

Reports

- **KPIs overview ---------- 8**
- **Loyalty & promotion impact ---------- 9**
- **Store & channel insights ---------- 11**
- **Customer segmentation ---------- 12**

Final dashboard ---------- 14

Top 3 recommendations ---------- 15

Drive link ---------- 17

Introduction

Project Overview:

Walmart is a global retail leader operating hypermarkets, discount stores, and e-commerce platforms. With rising competition from Amazon, Target, and other retailers, retaining customers has become a major challenge. Although Walmart collects large amounts of data, the current reports lack the depth needed to understand churn, loyalty behaviour, and promotional impact.

Project Objective

The primary objective of this project is to develop an interactive and data-driven Customer Retention Analytics Dashboard using Power BI. This dashboard aims to:

● Consolidate customer demographics, purchase history, store/e-commerce performance, and membership data.

● Enable dynamic segmentation of high-value, repeat, and churned customers.

● Provide actionable insights to improve retention, loyalty engagement, and regional strategies.

Dataset Description

1. **Customer_Demographics**
   Columns: Customer_ID, Age, Gender, Region, Income_Level, Membership_Since, Preferred_Channel (Store/Online)

2. **Customer_Transactions**
   Columns: Transaction_ID, Customer_ID, Store_ID, Product_Category (Groceries, Electronics, Apparel, Home & Living), Transaction_Date, Amount, Promotion_Applied (Yes/No)

3. **Store_Locations**
   Columns: Store_ID, Store_Type (Supercenter, Neighborhood Market, Sam's Club, Online), Region, Opening_Year

4. **Loyalty_Program**
   Columns: Customer_ID, Loyalty_Tier (Basic, Plus, Premium, Elite), Points_Earned, Points_Redeemed

5. **Churn_Labelled_Customers**
   Columns: Customer_ID, Last_Purchase_Date, Churn_Flag (0 = Active, 1 = Churned), Churn_Reason (Inactivity, Competitor, Low Engagement)

Data Cleaning & Modeling

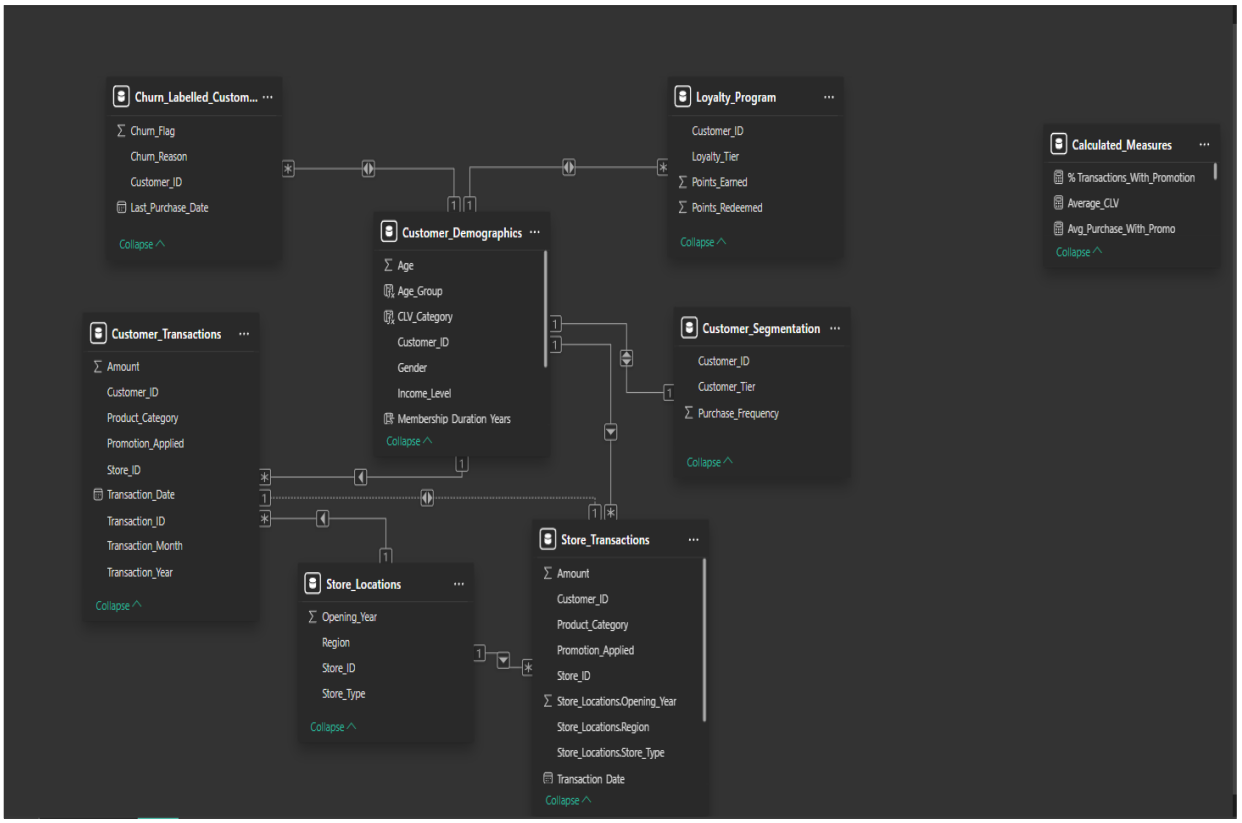
Power Query Steps

- Corrected data types
- Extracted date parts (Year, Month)

Calculated Columns

- **Membership Duration**
- **Age Group**
- **Purchase Frequency Segments**
- **CLV category**
- **Customer Tier**
- **Transaction Year**
- **Transaction Month**

Data Model Screenshot



DAX Measures Used

- **Total_Customers = COUNTROWS(Churn_Labelled_Customers)**

- **Churned_Customers =**
  CALCULATE(
  COUNTROWS(Churn_Labelled_Customers),
  FILTER(
  Churn_Labelled_Customers,
  Churn_Labelled_Customers[Churn_Flag] = 1
)
)

- **Churn_Rate = DIVIDE([Churned_Customers], [Total_Customers]) * 100**

- **Repeat_Customers =**
  CALCULATE(
  DISTINCTCOUNT(Customer_Transactions[Customer_ID]),
  FILTER(
  VALUES(Customer_Transactions[Customer_ID]),
  CALCULATE(COUNTROWS(Customer_Transactions)) > 1
)
)

- **Total_Transactions = COUNTROWS(Customer_Transactions)**

- **Transactions_With_Promotion =**
  CALCULATE(
  COUNTROWS(Customer_Transactions),
  Customer_Transactions[Promotion_Applied] = "Yes"
)

- **% Transactions_With_Promotion =**
  DIVIDE([Transactions_With_Promotion], [Total_Transactions]) * 100

- **Avg_Purchase_With_Promo =**
  CALCULATE(
  AVERAGE(Customer_Transactions[Amount]),
  Customer_Transactions[Promotion_Applied] = "Yes"
)

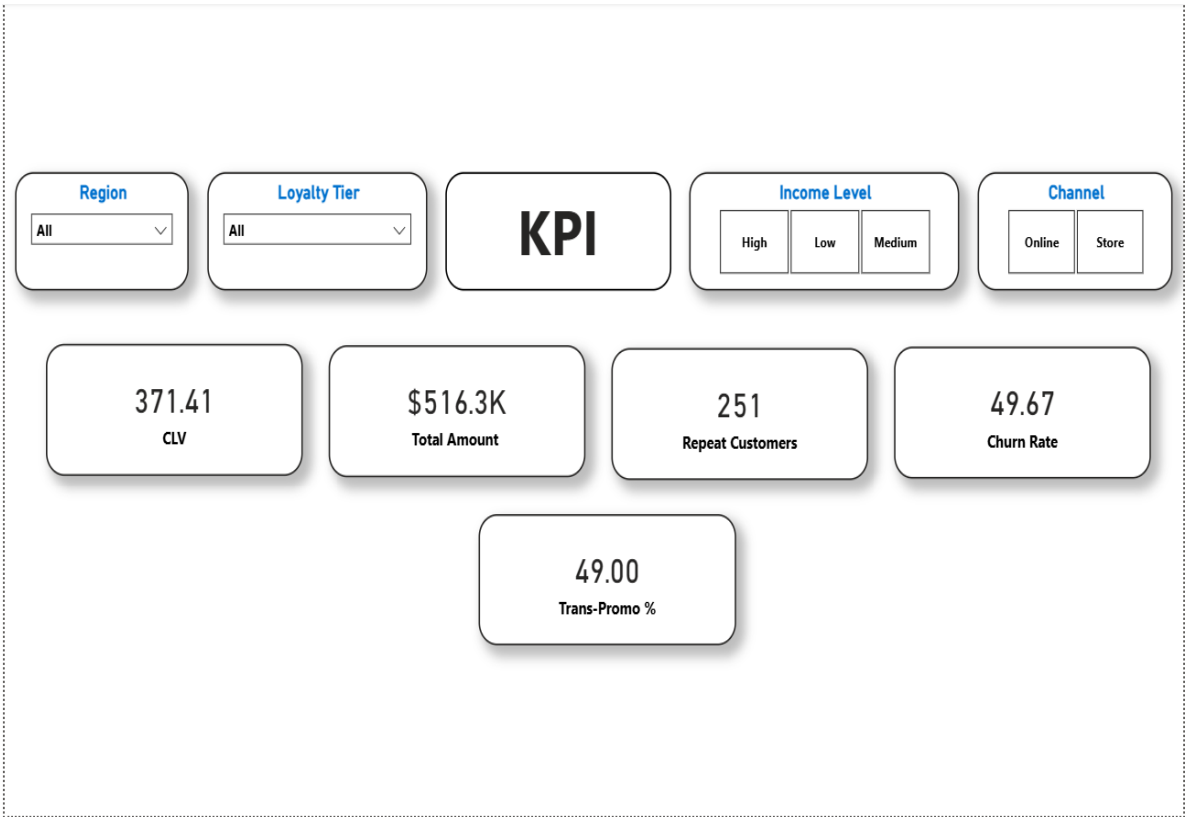
- **Avg_Purchase_Without_Promo =**
  CALCULATE(
  AVERAGE(Customer_Transactions[Amount]),
  Customer_Transactions[Promotion_Applied] = "No"
)

- **Total Amount Spent = SUM(Customer_Transactions[Amount])**

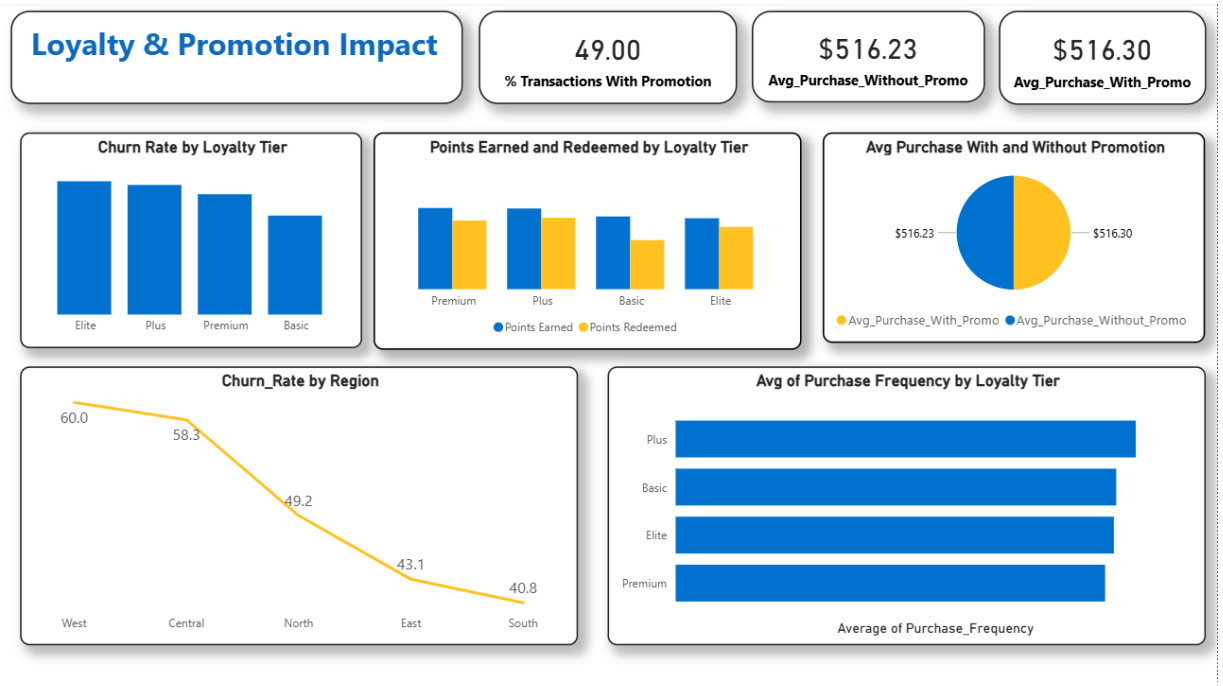
- **CLV =**
  DIVIDE(
  [Total Amount Spent],
  SUM(Customer_Demographics[Membership_Duration_Years])
)

- **Average_CLV =**
  AVERAGEX(
  ALL(Customer_Demographics),
  [CLV]
)

KPIs Overview



Loyalty & Promotion Impact



Insights:

Churn Rate by Loyalty Tier

- **Elite and Plus tiers have higher churn.**
- **Basic tier shows the lowest churn, as expectations are lower.**

Points Earned vs Redeemed

- **Premium and Plus customers earn many points but redeem fewer, showing poor loyalty program utilization.**
- **Basic customers show high redemption.**

Avg Purchase With vs Without Promotion

- **Purchases with promotions are slightly higher in value, confirming promotions positively influence spending.**
- **Nearly 49% of all transactions involve promotions.**

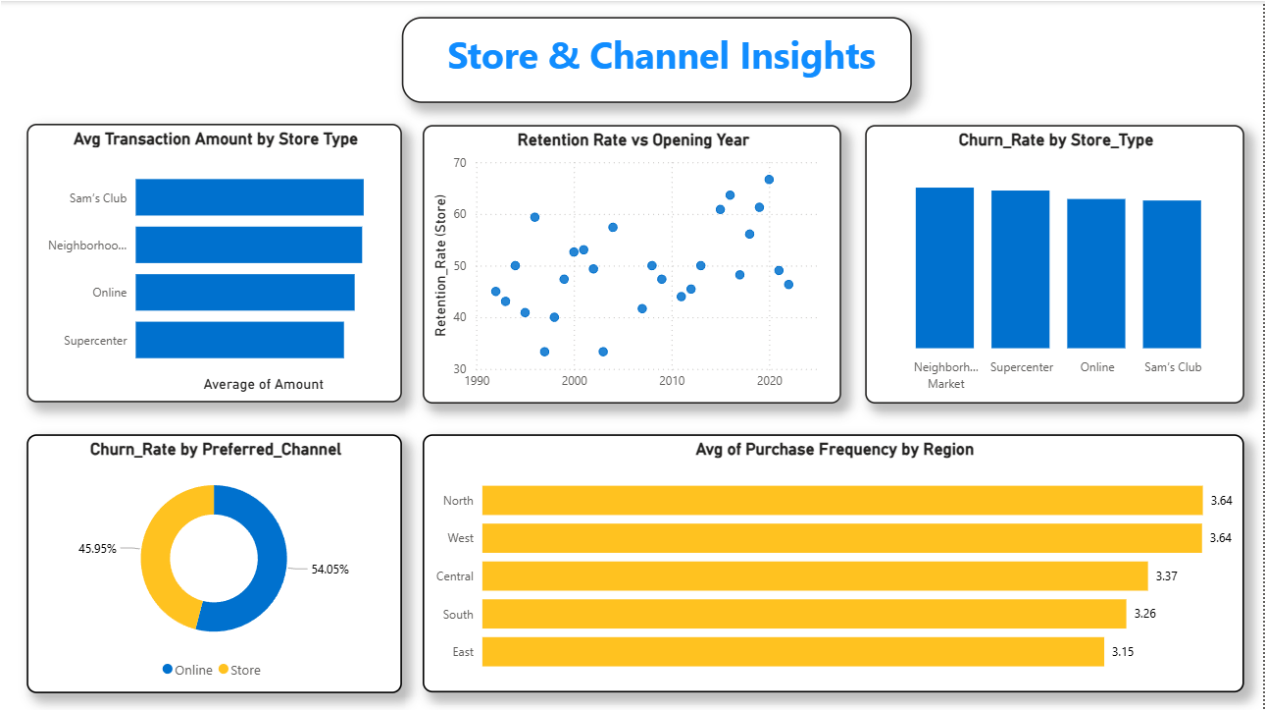
Churn Rate by Region

- **West and Central have the highest churn, indicating retention problems.**
- **South has the lowest churn, suggesting stronger customer experience.**

Avg Purchase Frequency by Loyalty Tier

- **Plus customers purchase most frequently, making them highly valuable.**
- **Basic and Elite have lower frequency than expected—Elite needs retention focus.**

Store & Channel Insights



Insights:

Average Transaction Amount by Store Type

- Sam's Club has the **highest average transaction amount**, indicating high-value shoppers.
- Online and Neighborhood Markets perform steadily with moderate amounts.

Retention Rate vs Store Opening Year

- No relationship exits.

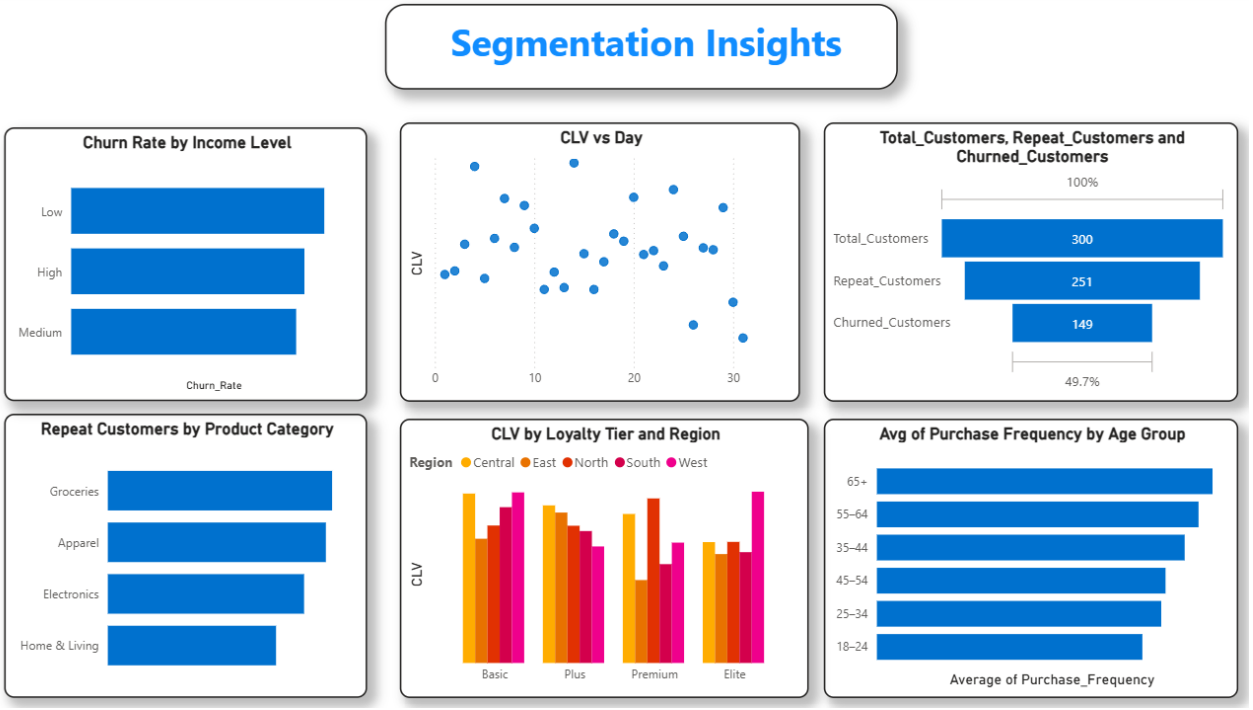
Churn Rate by Store Type

- Neighborhood Markets and Supercenters have slightly higher churn.
- Sam's Club has comparatively lower churn, showing stronger customer stickiness.

Average Purchase Frequency by Region

- North and West regions have the **highest purchase frequency** (3.64).
- East region has the lowest, indicating weaker engagement.

Customer Segmentation



Insights:

Churn Rate by Income Level

- **Low-income customers show slightly higher churn, indicating price sensitivity.**
- **High, Medium income groups also churn at similar levels, showing churn is broad-based.**

CLV vs Days Since Last Purchase

- **No relationship exits.**

Repeat Customers by Product Category

- **Groceries and Apparel have the highest repeat buyers, showing strong habitual purchasing.**
- **Home & Living has the lowest repeat rate, needing better engagement.**

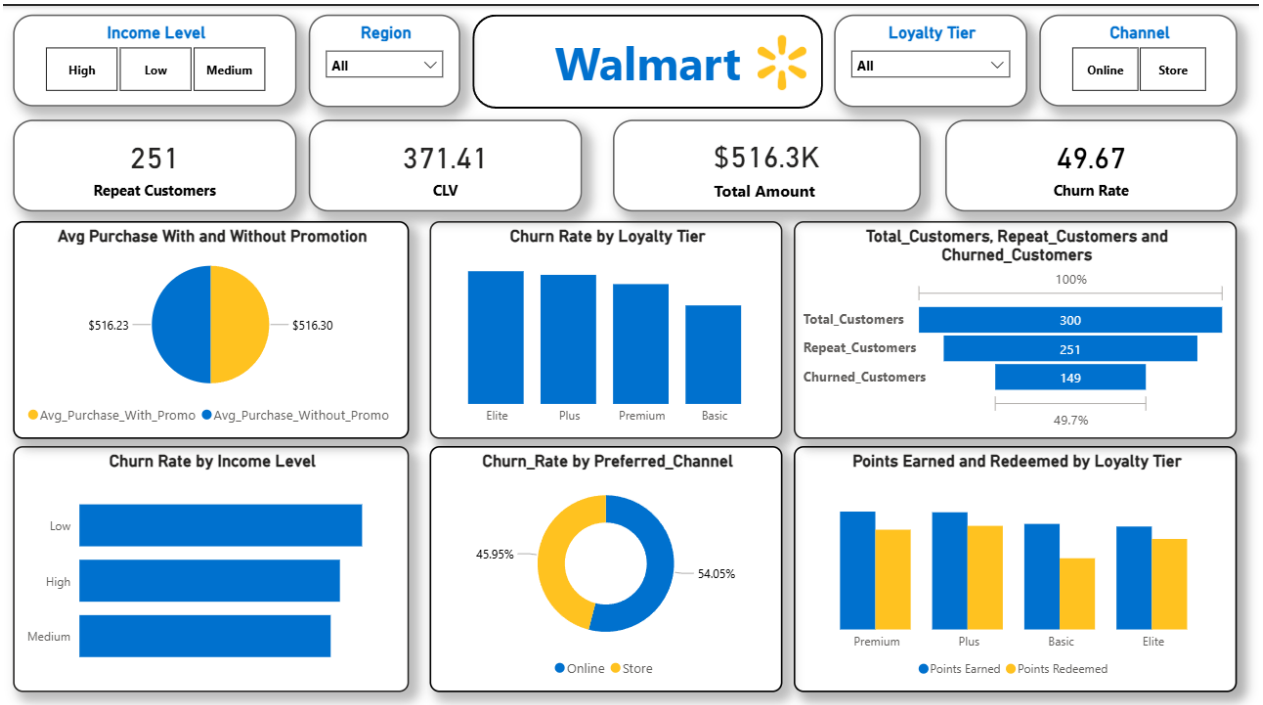
CLV by Loyalty Tier & Region

- **Premium customers show the highest CLV across most regions.**
- **West region consistently contributes the highest CLV across all tiers.**
- **Basic tier customers have lower CLV in every region.**

Average Purchase Frequency by Age Group

- **Age 65+ has the highest purchase frequency, showing strong loyalty.**
- **Young adults (18–24) have the lowest frequency, indicating low engagement.**

Final Dashboard



It summarizes all important insights from the analysis and presents them clearly to support better decisions on retention, promotions, and customer engagement.

Top 3 Recommendations:

Which customers to prioritize for retention?

Insight:
Plus tier customers show the highest purchase frequency, making them the most valuable for retention. Basic and Elite tiers have moderate frequency, indicating growth potential, whi le Premium customers show the lowest frequency, placing them at higher churn risk.

Recommendation:

Prioritize retention for:

- Plus tier: Maintain loyalty with personalized offers.
- Basic & Elite: Boost engagement with tier-upgrade rewards.
- Premium: Use targeted win-back offers to re-engage at-risk customers.

Which channels are underperforming?

Insight:

Churn rate is higher in the Online channel compared to the Offline channel.This indicates that online customers are less engaged, face more drop-
off points, or are not receiving enough value to stay active.

Recommendation:

To reduce online churn:

- Improve Online Experience: Faster checkout, better app/website UI.
- Offer Online-Exclusive Deals: Discounts, flash sales, or free delivery.
- Increase Engagement: Personalized notifications, reminders, and reward updates.
- Boost Loyalty Online: Extra bonus points for online purchases.

How to strengthen loyalty program engagement?

Insights:

- Elite & Plus have the highest churn, meaning these key tiers are at risk.
- Premium & Basic redeem fewer points, showing weak engagement.
- Promotions have little impact on purchase value (almost same avg).
- West & Central regions have the highest churn.
- Plus customers purchase most frequently, showing strong loyalty behavior.

Recommendations:

- Boost point values.

- Retain Elite & Plus with exclusive perks and targeted offers.
- Activate Premium & Basic using reminders for unused points and bonus-point campaigns.
- Fix high-churn regions with region-specific offers and better support.
- Optimize promotions—gives more offers for promotions.
- Add Gamification.

Drive Link

https://drive.google.com/file/d/1GwDDNBBWljfOvnhngkcYtJME_FjoHRtM/view?usp=drive_link